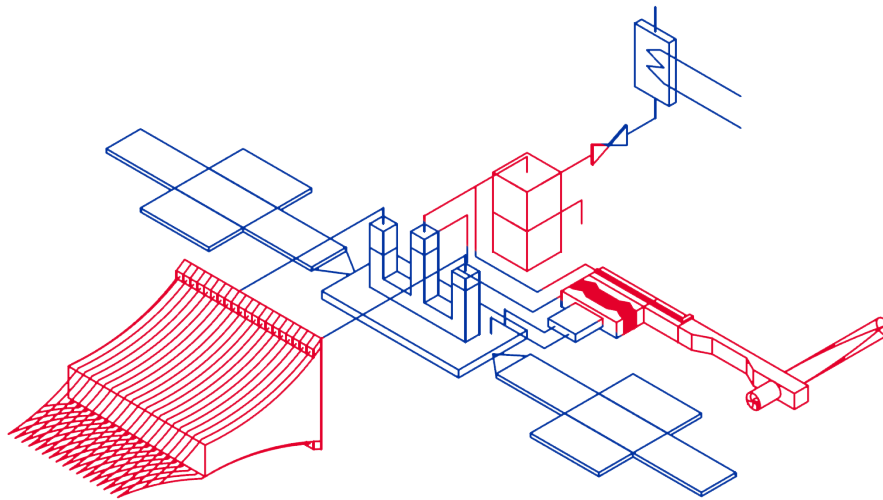


XSLR Aerospike Hybrid Expander Architecture, Thermodynamic Analysis, and Multi-Mission Scaling Study

Configurations A through D+, Option C Turbogenerator,
Bypass Management, Planetary Lander Study

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Document Summary

Engine denomination: XSLR — eXperimental aeroSpike Liquid Rocket
LOX/LH₂, Rocketdyne XRS-2200-derived geometry ($N_{\text{cell}} = 20$, $\varepsilon = 58$, $p_c = 50$ bar)

Final Architecture: Config. D+

Parallel full-expander dual-loop + Hybrid EEC + Turbogenerator

Thrust classes covered: 22 kN (demo), 40 kN (planetary lander), 220 kN (vehicle), 250 kN (extreme)

Key results: $I_{\text{sp,vac}} = 465$ s; throttle 5:1 (12–100 % RPL); battery 36 kg (22 kN) / 35 kg bootstrap (220 kN); $T_{\text{turb}} \leq 700$ K; DQ3 resolved; 16 open DQs consolidated

Companion reports: [1] • [2] • [3] • [5] • [4]

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Version 3.0 (consolidated)
Part of the Space Framework™.

Fact-checked: Sutton & Biblarz *RPE* 9th ed., Liang et al. *Acta Astronautica* 2021, Manski & Hagemann
Acta Astronautica 1997, NIST WebBook, NASA SP-8120, NASA CEA.



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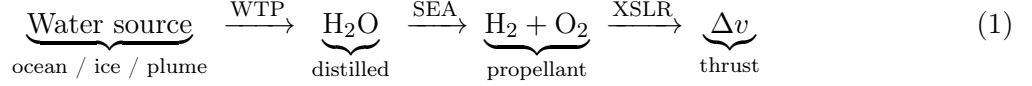
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1 Context and Motivation

1.1 XSLR within the Skyraker Architecture

The eXperimental aeroSpike Liquid Rocket (XSLR) is the chemical propulsion subsystem of Skyraker™, the propulsion heart of the Ikarus™ interplanetary spacecraft within the Space Framework™ architecture [1]. Skyraker integrates four subsystems in a closed-loop ISRU propulsion chain:



XSLR converts the electrolytically produced LH₂/LOX into high-thrust impulsive manoeuvres at $I_{\text{sp,vac}} = 465$ s and variable thrust from 12 to 100 % rated power level (RPL) in a single flight. The WTP [2], SEA [3], and ICER [5] subsystems are documented in companion reports.

1.2 Why a Linear Aerospike

The choice of a linear aerospike nozzle over a conventional bell nozzle is thermodynamically mandatory for the multi-mission requirement. At Titan’s surface pressure ($p_a = 146.7$ kPa), a bell nozzle at $\varepsilon = 58$ delivers $I_{\text{sp}} \approx 57$ s due to severe overexpansion losses — a Tsiolkovsky mass ratio of 95.6 for $\Delta V = 2550$ m/s, physically impossible for any vehicle. The altitude-compensating plug auto-matches the effective expansion ratio to ambient pressure, achieving ≈ 405 s and a mass ratio of 1.90 for the same mission. This single result drives the entire engine architecture choice.

1.3 Scope and Document Structure

This document consolidates all XSLR design studies conducted between January and June 2026 into a single self-contained reference. The structure follows the logical development: invariant parameters (§2) → configuration trade study (§3) → Config. D baseline at 22 kN (§4) → Config. D+ hybrid EEC at 22 kN (§5) → scaling to 220–250 kN and non-invariances (§6) → turbogenerator Option C (§7) → bypass management and DQ3 (§8) → in-orbit solar recharge (§9) → multi-mission applicability (§10) → consolidated open design questions (§11).

2 Engine Geometry and Cycle Invariants

2.1 XRS-2200 Heritage Geometry

All XSLR variants share the Rocketdyne XRS-2200 cell geometry [10]: $N_{\text{cell}} = 20$ cells in a linear array, area ratio $\varepsilon = 58$, plug half-angle $\delta = 20^\circ$, 80 % truncation factor. This geometry is flight-validated (X-33 programme) and provides well-characterised aerodynamic and thermal properties.

2.2 Thermodynamic Cycle Invariants & Geometric Scaling Laws

Table 1 collects all parameters that are independent of the thrust scaling factor $k = F_{\text{target}}/F_{\text{base}}$. These quantities do not change between the 22 kN demonstrator and the 250 kN flight engine.

With thrust scaling factor k :

$$A_t \propto k, \quad h_t, L_{\text{plug}}, W_{\text{span}} \propto \sqrt{k} \quad (2)$$

The throat heat flux from the calibrated Bartz correlation [11]:

$$q_0 = C_B p_c^{0.8} T_c^{0.68} \quad (3)$$

with $C_B = 3.220 \times 10^{-1} \text{ W m}^{-2} \text{ Pa}^{-0.8} \text{ K}^{-0.68}$ calibrated to the RS-25 reference point ($p_{c,\text{ref}} = 207$ bar, $T_{c,\text{ref}} = 3590$ K, $q_{0,\text{ref}} = 60 \text{ MW/m}^2$), giving $q_0 = 17.04 \text{ MW/m}^2$ at the XSLR design point. Because q_0 depends only on p_c and T_c , it is a **cycle invariant** and does not scale with thrust.

Table 1: Cycle-invariant design parameters, all XSLR variants.

Parameter	Symbol	Value
Chamber pressure	p_c	50 bar
Mixture ratio	O/F	4.8
Chamber temperature	T_c	3000 K
Characteristic velocity	c^*	2350 m/s
Vacuum specific impulse	$I_{sp,vac}$	465 s
Sea-level specific impulse	$I_{sp,SL}$	394 s
Turbine pressure ratio	Π_t	2.5
Turbine efficiency	η_t	0.72
Pump efficiency	η_p	0.70
Max turbine inlet temperature	$T_{turb,max}$	800 K
Plug half-angle	δ	20°
Plug truncation factor	f_{trunc}	0.80
Number of cells	N_{cell}	20
Tap-off bleed fraction	f_{bleed}	0 %

3 Configuration Trade Study

3.1 Two Cooling Surfaces and the Heat-Load Asymmetry

The linear aerospike presents two geometrically distinct heat-exchange surfaces, which become the two loops of the dual-loop architecture.

The **nozzle loop** covers the cell walls (throat and divergent section). Effective cooling area with Rao integral amplification factor $k_{Rao} = 4.5$:

$$A_{cool,n} = k_{Rao} A_t \quad (4)$$

The **spike loop** covers both faces of the plug body ($k_{spike} = 1.4$ for two-sided exposure):

$$A_{cool,s} = L_{plug} \times W_{span} \times k_{spike} \quad (5)$$

Using area-averaged flux ratios $\bar{\eta}_n = 0.55$ and $\bar{\eta}_s = 0.18$, and the 22 kN reference geometry ($A_t = 2.77 \times 10^{-3} \text{ m}^2$, $L_{plug} = 0.469 \text{ m}$, $W_{span} = 0.375 \text{ m}$):

$$\dot{Q}_n = q_0 A_{cool,n} \bar{\eta}_n = 17.04 \times 0.0125 \times 0.55 = 154 \text{ kW} \quad (6)$$

$$\dot{Q}_s = q_0 A_{cool,s} \bar{\eta}_s = 17.04 \times 0.246 \times 0.18 = 992 \text{ kW} \quad (7)$$

$$\frac{\dot{Q}_s}{\dot{Q}_n} = 6.44 \quad (8)$$

The spike loop absorbs 6.44× more thermal energy than the nozzle loop at 22 kN. This fundamental asymmetry motivates the dual-loop architecture: the dominant heat source (spike) powers the dominant turbine.

3.2 Four Candidate Configurations

Four architectures are evaluated against figures of merit weighted for the HTHL SSTO mission profile.

Config. A (single-loop tap-off reference): a single LH₂ loop on the nozzle cells. Turbopumps driven by 5.5 % hot-gas bleed at $T_{\text{turb}} = 1050$ K. $I_{\text{sp,vac}} \approx 460$ s (bleed-corrected). Baseline.

Config. B (parallel dual-loop tap-off): two independent LH₂ loops in parallel, tap fraction reduced to 2.5 %; turbine temperature $T_{\text{turb}} = 800$ K. Modest I_{sp} gain; Mode-2 available in limited form.

Config. C (series dual-loop tap-off): nozzle and spike loops in series; simplest hardware change from Config. A. Mode-2 requires an additional diverter valve; lowest complexity score.

Config. D (parallel full expander, no tap-off): two completely independent expander loops, each with its own turbine, zero bleed penalty ($\delta I_{\text{sp}} \approx +5$ s over Config. A), $T_{\text{turb}} \leq 800$ K, inherent Mode-2 thrust modulation via 8+12 cell split ($\varepsilon_{\text{eff}} : 58 \rightarrow 97$, $\Delta F \approx -40$ %). Pure-expander throttle floor at ≈ 35 % RPL, difficult cold start (igniters needed).

Config. D+ (hybrid EEC, recommended): extends Config. D by replacing the spike-loop turbopump with a BLDC motor; 5:1 throttle, combined Mode-2 + throttle (12–100 % RPL), cold start without igniters.

3.3 HTHL-Weighted Trade Matrix

Table 2: Configuration trade matrix, 22 kN, HTHL-weighted scores (0–1, higher is better). Weights: $I_{\text{sp,vac}}$ (0.30), T_{turb} (0.25), thrust modulation (0.20), Π_W (0.15), complexity (0.10).

Config.	I_{sp}	T_{turb}	Thrust mod.	Π_W	Complexity	Total
A (tap-off ref.)	0.000	0.000	0.00	0.00	1.00	0.100
B (par. dual tap)	0.000	1.000	0.15	1.00	0.50	0.498
C (series tap)	0.000	1.000	0.00	0.87	1.00	0.381
D (full exp.)	1.000	1.000	0.50	0.47	0.00	0.671
D+ (EEC)	1.000	1.000	1.00	0.60	0.00	0.760

Config. D+ scores 0.760, driven uniquely by its combined Mode-2 + continuous throttle capability, which is unavailable in any other configuration. The series vs. parallel comparison shows that parallel is mandatory whenever Mode-2 operation is required, as series architecture cannot shut down the nozzle section without starving the spike section.

4 Configuration D: Parallel Full-Expander Baseline (22 kN)

4.1 Propellant Flows and Engine Geometry

At sea-level thrust $F_{\text{SL}} = 22\,100$ N, $I_{\text{sp,SL}} = 394$ s:

$$\dot{m}_{\text{eng}} = \frac{F_{\text{SL}}}{I_{\text{sp,SL}} g_0} = \frac{22100}{394 \times 9.807} = 5.90 \text{ kg/s} \quad (9)$$

$$\dot{m}_{\text{H}_2} = \frac{\dot{m}_{\text{eng}}}{1 + O/F} = 1.017 \text{ kg/s}, \quad \dot{m}_{\text{LOX}} = 4.880 \text{ kg/s} \quad (10)$$

Total throat area:

$$A_t = \frac{\dot{m}_{\text{eng}} c^*}{p_c} = \frac{5.90 \times 2350}{50 \times 10^5} = 2.77 \times 10^{-3} \text{ m}^2 \quad (11)$$

Cell throat half-height (using XRS-2200 proportional relation $A_{t,\text{cell}} = 2.5 h_t^2$):

$$h_t = \sqrt{A_{t,\text{cell}}/2.5} = \sqrt{(2.77 \times 10^{-3}/20)/2.5} = 7.50 \text{ mm} \quad (12)$$

Truncated plug length ($\delta = 20^\circ$, $f_{\text{trunc}} = 0.80$):

$$L_{\text{plug}} = \frac{(\varepsilon - 1) h_t}{2 \tan \delta} f_{\text{trunc}} = \frac{57 \times 7.50}{2 \tan 20^\circ} \times 0.80 = 37.6 \text{ cm} \quad (13)$$

$$W_{\text{span}} = N_{\text{cell}} \times 5 h_t = 20 \times 5 \times 7.50 = 37.5 \text{ cm} \quad (14)$$

4.2 Coolant Flow Requirements

Minimum LH₂ mass flow per loop to satisfy the Inconel 718 wall temperature limit ($T_{\text{wall,max}} = 900 \text{ K}$, $\Delta T_{\text{max}} = 740 \text{ K}$, $\bar{c}_{p,\text{H}_2} = 9500 \text{ J/(kg K)}$):

$$\dot{m}_{\text{H}_2,\text{n,min}} = \frac{\dot{Q}_n}{\bar{c}_{p,\text{H}_2} \Delta T_{\text{max}}} = \frac{154\,000}{9500 \times 740} = 0.022 \text{ kg/s} \quad (15)$$

$$\dot{m}_{\text{H}_2,\text{s,min}} = \frac{\dot{Q}_s}{\bar{c}_{p,\text{H}_2} \Delta T_{\text{max}}} = \frac{992\,000}{9500 \times 740} = 0.141 \text{ kg/s} \quad (16)$$

The combined minimum cooling flow of 0.163 kg/s is $\approx 16\%$ of total fuel flow; the remainder bypasses directly to the injectors.

4.3 Turbine Power and Pump Demand

Isentropic expansion efficiency ($\gamma_{\text{H}_2} = 1.40$, $\Pi_t = 2.5$):

$$\eta_{\text{exp}} = 1 - \Pi_t^{-(\gamma-1)/\gamma} = 1 - 2.5^{-0.286} = 0.230 \quad (17)$$

Both loops operate at $T_{\text{out}} = 800 \text{ K}$, $c_{p,\text{turb}} = 14\,300 \text{ J/(kg K)}$:

$$\dot{W}_{\text{turb,s}} = 0.141 \times 14\,300 \times 800 \times 0.72 \times 0.230 = 257 \text{ kW} \quad (18)$$

$$\dot{W}_{\text{turb,n}} = 0.022 \times 14\,300 \times 800 \times 0.72 \times 0.230 = 40 \text{ kW} \quad (19)$$

Pump demands ($\rho_{\text{LH}_2} = 71 \text{ kg/m}^3$, $\rho_{\text{LOX}} = 1141 \text{ kg/m}^3$, $\Delta p = 46 \text{ bar}$):

$$\dot{W}_{\text{pump,H}_2} = \frac{1.017 \times 46 \times 10^5}{71 \times 0.70} = 94 \text{ kW} \quad (20)$$

$$\dot{W}_{\text{pump,LOX}} = \frac{4.880 \times 46 \times 10^5}{1141 \times 0.70} = 28 \text{ kW} \quad (21)$$

Power ratio confirms cycle self-sustainability:

$$\Pi_W = \frac{\dot{W}_{\text{turb,total}}}{\dot{W}_{\text{pump,total}}} = \frac{297}{122} = 2.43 \quad (> 1 \checkmark) \quad (22)$$

4.4 Mode-2 Thrust Modulation and Cell Assignment

The 20 cells are split: 8 inner cells assigned to the nozzle loop (Mode-1 only), 12 outer cells to the spike loop (Modes 1 and 2). On nozzle-loop shutdown (Mode-2 entry):

$$\varepsilon_{\text{eff,M2}} = \frac{N_{\text{cell}}}{N_{\text{spike}}} \times \varepsilon = \frac{20}{12} \times 58 = 96.7 \quad (23)$$

$$I_{\text{sp,vac,M2}} \approx 465 \times \left(1 + 0.015 \ln \frac{96.7}{58} \right) = 468.6 \text{ s} \quad (24)$$

The I_{sp} gain in Mode-2 is modest (+3.6 s); the primary function is $\approx -40\%$ thrust reduction for structural load management during the high-dynamic-pressure HTHL ascent phase. The I_{sp} gain from eliminating the tap-off bleed vs. Config. A:

$$\delta I_{\text{sp}} = f_{\text{tap}}(I_{\text{sp,vac}} - I_{\text{sp,dump}}) = 0.055 \times (465 - 195) = 14.85 \text{ s} \quad (25)$$

giving Config. D an effective $I_{\text{sp,vac}} \approx 465 \text{ s}$ vs. $\approx 450 \text{ s}$ for Config. A.

Primary weakness of Config. D: The throttle floor lies at $\approx 35\%$ RPL ($\Pi_W \rightarrow 1$); below this the cycle collapses. Combined Mode-2 + continuous throttle is not available. Both limitations are eliminated in Config. D+.

5 Configuration D+: Hybrid Electric Expander Cycle (22 kN)

5.1 Core Architectural Modification

Config. D+ applies the electric expander cycle (EEC) principle of Liang et al. [7] to the dual-loop aerospike. Two changes from Config. D:

1. The spike-loop LH_2 pump (Pump S, $\dot{W} \approx 94 \text{ kW}$) is replaced by a BLDC motor-driven centrifugal pump powered by a lithium-polymer battery. The spike-loop turbine is retained but now drives the LOX pump as auxiliary.
2. The nozzle-loop turbine becomes the primary LOX pump driver (margin $\Pi_{W,N} = 1.61$); the spike-loop turbine provides supplementary LOX assistance via a sprag-clutch freewheel.

Four control variables define the system state: P_{el} (primary thrust input), α (flow-split ratio), β_N (nozzle turbine bypass), β_{LOX} (LOX pump bypass).

Structural analogy with Liang et al. EEC (LOX/LCH4): LCH4 pump $\rightarrow$ spike-loop LH_2 pump (electric); turbine drives LOX pump $\rightarrow$ nozzle-loop turbine drives LOX pump; single cooling channel $\rightarrow$ spike cooling path (dominant heat surface); battery: 50.65 kg (Liang) $\rightarrow$ 36 kg (XSLR, smaller due to higher turbine offload).

5.2 Power Balance at 22 kN, 100 % RPL

Design-point split $\alpha = 0.25$:

$$\dot{m}_{\text{H}_2, \text{s}} = 0.254 \text{ kg/s}, \quad \dot{m}_{\text{H}_2, \text{n}} = 0.763 \text{ kg/s} \quad (26)$$

Electric pump demand ($\Delta p_{\text{pump}, \text{S}} = p_c + \Delta p_{\text{chan}} + \Delta p_{\text{inj}} = 55.5 \text{ bar}$):

$$\dot{W}_{\text{pump}, \text{S}} = \frac{\dot{m}_{\text{H}_2, \text{s}} \Delta p_{\text{pump}, \text{S}}}{\rho_{\text{LH}_2} \eta_p} = \frac{0.254 \times 55.5 \times 10^5}{71 \times 0.70} = 28.4 \text{ kW} \quad (27)$$

Battery draw (motor $\eta_m = 0.90$, inverter $\eta_{\text{inv}} = 0.92$):

$$P_{\text{batt}} = \frac{\dot{W}_{\text{pump}, \text{S}}}{\eta_p \eta_m \eta_{\text{inv}}} = 49 \text{ kW} \quad (28)$$

Nozzle-loop partial routing for turbine (target $T_{\text{out}, \text{N}} = 700 \text{ K}$):

$$f_N = \frac{\dot{Q}_n}{(700 - 60) \dot{m}_{\text{H}_2, \text{n}} \bar{c}_p} = \frac{154\,000}{640 \times 0.763 \times 9500} = 0.033 \quad (29)$$

$$\dot{W}_{\text{turb}, \text{N}} = 0.0253 \times 14300 \times 700 \times 0.72 \times 0.222 = 45 \text{ kW} \quad \Pi_{W, \text{N}} = 45/28 = 1.61 \checkmark \quad (30)$$

Table 3: Power budget comparison: Config. D vs. Config. D+ at 100 % RPL, 22 kN.

Item	Config. D	Config. D+	Note
$\dot{W}_{\text{turb,s}}$ available	+257 kW	+257 kW	unchanged
$\dot{W}_{\text{turb,n}}$ available	+40 kW	+45 kW	partial routing
$\dot{W}_{\text{pump,H}_2,\text{s}}$	−94 kW	0 kW	replaced by motor
$\dot{W}_{\text{pump,H}_2,\text{n}}$	−28 kW	−28 kW	turbopump
$\dot{W}_{\text{pump,LOX}}$	−28 kW	−28 kW	turbine N shaft
P_{batt} required	~5 kW	−49 kW	spike pump only
Turbine surplus / bypass	+147 kW	+246 kW	H ₂ bypass inj.
Net balance	+152 kW	+197 kW	> 0 ✓

5.3 Three Operating Modes

Table 4: Config. D+ operating modes, 22 kN engine.

Mode	RPL (%)	p_c (bar)	F (N)	P_{batt} (kW)	Turb. frac. (%)	$I_{\text{sp,vac}}$ (s)	$T_{\text{turb,N}}$ (K)
1 Boost	100	50	22 100	49	31	465	700
2 Cruise	60	30	13 260	18	48	462	590
3 Throttle	20	10	4 420	10	3	455	380
Ratio	5:1	5:1	5:1	4.9:1	—	—	—

5.4 Battery Mass Estimate

Following the Liang et al. method [7]:

$$m_{\text{batt}} = k_b \max\left(\frac{P_{\text{batt}}}{\xi_b}, \frac{W_l}{\tau_b}\right) \quad (31)$$

with NMC lithium-ion parameters $\xi_b = 6 \text{ kW/kg}$, $\tau_b = 468 \text{ kJ/kg}$, safety factor $k_b = 1.10$. For the HTHL five-phase mission profile ($W_l = 15\,120 \text{ kJ}$):

$$m_{\text{batt,P}} = 9.0 \text{ kg}, \quad m_{\text{batt,E}} = 35.5 \text{ kg} \quad (32)$$

The energy constraint governs: $m_{\text{batt}} \approx \mathbf{36 \text{ kg per engine}}$ (30.8 % of a fully electric-pump equivalent at 117 kg; consistent with Liang et al.’s 64.1 % ratio for their smaller engine, the difference reflecting the higher turbine offload fraction of the aerospike spike loop).

5.5 Throttle Performance and Turbine Efficiency

Turbine efficiency follows the Liang et al. impulse-turbine model [7]:

$$\eta_t = \alpha_1 \left(\frac{u}{C_{\text{ad}}}\right)^2 + \alpha_2 \left(\frac{u}{C_{\text{ad}}}\right) + \alpha_3 \quad (33)$$

with coefficients $\alpha_1 = -0.551$, $\alpha_2 = 1.778$, $\alpha_3 = 0.002$ (optimum $u/C_{\text{ad}} \approx 0.48$, $\eta_t \approx 0.43$). In a fixed-speed turbopump at 20 % RPL, C_{ad} drops $\approx 40 \%$, shifting $u/C_{\text{ad}} \rightarrow 0.80$ and $\eta_t \rightarrow 0.22$ (−51 %

— the Liang et al. result). In Config. D+, a variable-speed LOX pump maintains $u/C_{ad} \approx 0.48$ across the full throttle range, preserving $\eta_t \approx 0.43$ throughout. The 5:1 throttle ratio is verified at 20 % RPL: minimum cooling flow ($0.141 \times 0.20 = 0.028$ kg/s), $P_{el,min} \approx 10$ kW, $p_c \approx 10$ bar ($> p_{c,min} = 6$ bar for pintle injectors) — all constraints satisfied.

5.6 Cold Start, Restart, and Mode-2 with Continuous Throttle

Start sequence: T–30 s: pre-chill lines; T–5 s: BLDC motor spin-up, LH₂ enters spike channels at 10 % flow; T0: ignition, chamber reaches $T_c \approx 3000$ K within 0.8 s; T+2 s: nozzle-loop turbine at operating speed, takes over LOX pump; T+3 s: steady state. No pyrotechnic igniters required; target restart count: > 100 without maintenance.

With combined Mode-2 cell shutdown ($\varepsilon_{eff} : 58 \rightarrow 97$) and electric-pump throttle, the effective thrust envelope spans **12–100 % RPL continuously** — a capability unavailable in Config. D.

6 Scaling to 220–250 kN: Non-Invariances

6.1 Scaling Laws and Power Budget at 220 kN

Scaling factor $k = 10$ (220 kN/22 kN). All extensive quantities scale as k ; all thermodynamic intensive quantities are cycle invariants (Table 1).

Table 5: XSLR parameters across thrust classes. Cycle-invariant quantities in italics.

Parameter	XSLR-22	XSLR-40	XSLR-220	XSLR-250
k (scaling factor)	1.00	1.81	10.00	11.31
$\dot{m}_{eng}$ (kg/s)	5.90	10.35	59.00	64.70
A_t (cm ²)	2.77	5.01	27.70	304.1
h_t (mm)	7.50	9.87	23.73	24.66
L_{plug} (cm)	37.6	61.8	118.9	154.5
W_{span} (cm)	37.5	98.6	118.6	246.6
$I_{sp,vac}$ (s)	<i>465</i>	<i>465</i>	<i>465</i>	<i>465</i>
q_0 (MW/m ²)	<i>17.04</i>	<i>17.04</i>	<i>17.04</i>	<i>17.04</i>
$\dot{Q}_s/\dot{Q}_n$	6.44	12.76	6.44	12.76
Π_W (Config. D pure)	1.86	3.48	1.86	3.48
$\dot{W}_{bypass}$ (kW)	246	543	2065	3393
Bypass fraction (%)	51	71	80	71
Battery / bootstrap (kg)	36	36	35	35

6.2 Three Critical Non-Invariances

Three quantities commonly assumed to be cycle-invariant break at large k . These are the most important analytical findings of the scaling exercise.

Non-invariance 1 — heat-load ratio. At 22 kN: $\dot{Q}_s/\dot{Q}_n = 6.44$. At 250 kN, the longer plug in absolute terms produces a larger exhaust recirculation zone and higher effective $\bar{\eta}_s$:

$$\left. \frac{\dot{Q}_s}{\dot{Q}_n} \right|_{250 \text{ kN}} = 12.76 \quad (+98 \%) \quad (34)$$

The minimum spike cooling fraction rises from $\alpha_{min} = 13.9\%$ to 20.9% .

7.5 Option C Trade Matrix

Table 6: Power architecture trade at 220 kN. Scores normalised 0–1, higher is better.

Option	I_{sp} (0.25)	Throttle (0.20)	Mission (0.20)	Battery (0.15)	Start (0.10)	Bypass (0.10)	Total
A (pure D)	1.00	0.60	1.00	1.00	0.30	0.40	0.774
B (EEC+batt.)	1.00	1.00	0.50	0.25	1.00	0.40	0.728
C (turbogen.)	1.00	1.00	1.00	0.85	0.95	0.50	0.921
D (solar+EEC)	1.00	1.00	0.70	0.45	0.95	0.40	0.780

Option C (turbogenerator) is the recommended architecture for all multi-mission XSLR variants above 40 kN. It simultaneously resolves the battery mass bottleneck, preserves full 5:1 throttle and Mode-2 capability, and converts bypass surplus into useful electrical work.

8 Bypass Thermal Management and DQ3

8.1 Bypass Residual at 220 kN

Under Option C, 2065 kW of pre-heated H_2 ($T = 800$ K, $p = 21$ bar, $\dot{m}_{bypass} = 1.13$ kg/s, 11.1 % of total LH_2) remains after generator extraction. Three recovery strategies are evaluated.

S1 — Recuperative pre-heater (HX-R). A counter-flow heat exchanger transfers enthalpy from the bypass stream to the cold LH_2 feed before the flow-split valve. Setting bypass outlet at 480 K:

$$\dot{Q}_{HXR} = \dot{m}_{bypass} c_{p,gas} (800 - 480) = 1.13 \times 14300 \times 320 \approx 5170 \text{ kW} \quad (40)$$

LH_2 feed pre-heated by ≈ 54 K; bypass injected at 480 K. Mass penalty: ≈ 8 kg. No new rotating hardware.

S2 — Bypass turbine stage with auxiliary generator. A second impulse turbine extracts an additional 630 kW from the bypass stream, reducing injection temperature to ≈ 680 K. Mass penalty: ≈ 25 kg; suitable as complement, not standalone.

S3 — Helical heat exchanger into nozzle loop (HX-N). The bypass stream heats *only* the nozzle-loop LH_2 feed before the nozzle cooling channels, raising its inlet temperature to 120 K and reducing bypass injection temperature to ≈ 350 K. Nozzle turbine power margin improves from 1.6 to 1.8. Mass penalty: ≈ 12 kg.

Recommended: S1 + S3 combined. HX-N (S3) cools bypass 800 K $\rightarrow$ ≈ 530 K; HX-R (S1) brings it further to ≈ 350 K. Total mass addition: 20 kg. Overall LH_2 feed pre-heat: ≈ 90 K. Bypass injection temperature 350 K eliminates the O/F stratification risk at the injector face. Combined with S3, DQ4 is reduced from HIGH to MEDIUM risk.

8.2 DQ3: Deep-Throttle Generator-to-Battery Transition

The crossover RPL at which generator output equals pump shaft demand:

$$RPL^* = \frac{\dot{W}_{pump,s,shaft}}{\dot{W}_{gen,100\%}} = \frac{284}{505} = 56.3\% \quad (41)$$

Above RPL^* the generator is in surplus; below it the battery supplements. A bidirectional bus voltage regulator (BVR) with 10 kHz switching bandwidth (GaN MOSFET, 4 kg, co-packaged with

existing inverter) maintains $V_{\text{bus}} = 800 \text{ V}$ within $\pm 5 \text{ V}$ across any RPL transition — four orders of magnitude faster than the $\sim 1 \text{ Hz}$ throttle ramp. Bus voltage deviation during transition: $< 0.6 \%$; transition completion time: $< 0.1 \text{ s}$.

DQ3 is fully resolved by the dual-mode BVR architecture. No firmware changes are required to the pump load controller; it sees a stable 800 V bus regardless of power source.

9 In-Orbit Solar Recharge

The 35 kg bootstrap battery (stored energy $\approx 14\,900 \text{ kJ}$) is recharged between burns by photovoltaic arrays. Recharge time:

$$t_{\text{recharge}} = \frac{W_{\text{stored}}}{\eta_{\text{charge}} P_{\text{solar}}} = \frac{14\,900 \times 10^3}{0.95 P_{\text{solar}}} \quad (42)$$

Table 7: Solar array comparison for XSLR in-orbit recharge of the 35 kg bootstrap battery. LILT = Low-Intensity Low-Temperature qualification.

Technology	Area (m ²)	Mass (kg)	$P_{1 \text{ AU}}$ (kW)	ξ (W/kg)	LILT qual.	$t_{\text{rech,Earth}}$ (h)	$t_{\text{rech,Jup.}}$ (h)
ROSA $\times 4$ units	56	188	21.6	115	No *	0.20	5.18
Juno $\times 3$ wings	72	340	14.0	41	Yes	0.30	8.45
Psyche $\times 2$ wings	75	285	20.0	70	Yes	0.21	6.00

*ROSA flexible GaAs cells show 15–25 % V_{OC} degradation below -20°C (NASA Glenn, 2017); qualification beyond 2 AU is not established.

Recommended: Psyche $\times 2$ wings for missions to Jupiter and inner targets. Best combination of specific power (70 W/kg vs. 41 W/kg for Juno), LILT qualification (screened ZTJ cells, verified to 2.9 AU), and recharge time (6.0 h at Jupiter orbit, well within 24 h ops ceiling). For Saturn-class operations (Titan, Enceladus), where recharge time reaches 19.7 h, an MMRTG ($\sim 3 \text{ kW}$) reduces it to $< 5 \text{ h}$ and is recommended as a mission-level supplement (DQ8).

10 Multi-Mission Applicability Across Thrust Classes

10.1 The Aerospike Advantage at Titan (Mission-Enabling)

A conventional bell nozzle with $\varepsilon = 58$ at 250 kN delivers at Titan’s surface ($p_a = 146.7 \text{ kPa}$, $A_e = 1.764 \text{ m}^2$):

$$F_{\text{bell,Titan}} = F_{\text{vac}} - p_a A_e = 295 \text{ kN} - 146\,700 \times 1.764 = 36.3 \text{ kN} \quad (43)$$

$$I_{\text{sp,bell,Titan}} = \frac{36\,300}{64.7 \times 9.807} = 57 \text{ s} \quad (44)$$

Tsiolkovsky mass ratio for $\Delta V = 2550 \text{ m/s}$ at 57 s: $m_0/m_f = e^{2550/(57 \times 9.807)} = 95.6$ — physically impossible. The aerospike auto-matches ε_{eff} to p_a , achieving $\eta_{\text{CF}} = 0.87$ [12]:

$$I_{\text{sp,aero,Titan}} = 0.87 \times 465 = 404.6 \text{ s} \quad \Rightarrow \quad m_0/m_f = 1.90 \quad (45)$$

Propellant fraction 47.4 % — achievable for a 3000 kg lander.

10.2 XSLR-40: Planetary Lander Thrust Class

The Venus 56 km altitude environment ($g = 8.87 \text{ m/s}^2$, $p_a = 53.1 \text{ kPa}$, $\eta_{\text{CF}} = 0.93$) drives the minimum thrust for a 3000 kg lander at $T/W \geq 1.3$:

$$F_{\text{vac,min}} = \frac{1.3 \times 3000 \times 8.87}{0.93} = 37.2 \text{ kN} \quad \Rightarrow \quad F_{\text{design}} = 40 \text{ kN} \quad (46)$$

Table 8: XSLR-40 per-target performance, $m_0 = 3000 \text{ kg}$, Config. D+ ($k = 1.81$).

Target	p_a (Pa)	η_{CF}	I_{sp} (s)	F (kN)	T/W @ 100 %RPL
Titan surface	146700	0.87	404.6	41.1	10.1
Venus 56 km	53100	0.93	432.5	43.9	1.65
Enceladus	0.1	1.00	465.0	47.2	139
Vacuum (ref.)	0	1.00	465.0	47.2	—

Table 9: Round-trip propellant budget, XSLR-40, $m_0 = 3000 \text{ kg}$. Titan is the mission design driver.

Target	ΔV (m/s)	I_{sp} (s)	m_0/m_f	m_{prop} (kg)	m_{H_2} (kg)	m_{LOX} (kg)
Titan	2550	404.6	1.90	1422	245	1177
Venus	1050	432.5	1.28	658	113	545
Enceladus	585	465.0	1.14	361	62	299

Titan requires throttle ratio $\geq 7.2:1$ for precision landing ($T/W_{\text{max}} = 10.1$ at 100 % RPL, target $T/W = 1.2$ at minimum):

$$\text{TR}_{\text{Titan}} = T/W_{\text{max}}/T/W_{\text{min}} = 10.1/1.20 = 8.4 \quad (47)$$

An Extended D+ with variable-area pintle injector targeting 8:1 is the required architecture (DQ-L1, §11). Venus hovers at $\geq 50 \%$ RPL only (below that $F < m_0 g_{\text{Venus}}$).

10.3 XSLR-250: Non-Invariances at Extreme Scale

At $k = 11.31$ (250 kN) the three non-invariances of §6.2 are maximally pronounced: $\dot{Q}_s/\dot{Q}_n = 12.76$, $\Pi_W = 3.48$, $\dot{W}_{\text{bypass}} = 3393 \text{ kW}$ (71.2 % of turbine power). The analysis confirms that Option C with S1+S3 bypass management is necessary at all thrust classes above $\sim 40 \text{ kN}$. Config. D at 250 kN is a valid intermediate hot-fire data step for DQ-A and DQ-D before committing to Option C hardware; it is not the flight architecture.

10.4 Target Environments and Material Constraints

Table 10: Target environments for XSLR-40 multi-body lander.

Target	g (m/s ²)	p_{surf} (kPa)	T_{surf} (K)	Critical constraint
Titan	1.352	146.7	93.7	Cryogenic restart; CH ₄ lakes
Enceladus	0.113	≤ 0.0001	75	Coldest target; H ₂ O plumes
Venus 56 km	8.87	53.1	272	H ₂ SO ₄ clouds; 100 m/s winds

11 Consolidated Open Design Questions

Table 11: Consolidated XSLR open design questions.

ID	Risk	Description
DQ-A	HIGH	Actual $\dot{Q}_s$ distribution under base-bleed injection; 71 % bypass surplus depends critically on this value. CFD and hot-fire survey mandatory before any Phase 2 hardware.
DQ-D	HIGH	Bypass injector combustion stability at 71 % bypass fraction; no precedent above $\sim 10\%$. Risk reduced to MEDIUM after S1+S3 (350 K injection); hot-fire validation at $\geq 50\%$ fraction still required.
DQ-L1	HIGH	XSLR-40 throttle ratio 8:1; variable-area pintle injector beyond Liang et al. 5:1 demonstration; combustion stability at 12.5 % RPL (Titan precision landing) unvalidated.
DQ-B	MED	Spike channel pressure drop at design flow; $\Delta p_{\text{chan}} = 3.5$ bar assumed; controls pump sizing and Π_W .
DQ-C	MED	LH ₂ supercritical transition at $T_{\text{out,s}} = 735$ K, $p_{\text{chan}} \approx 54$ bar; local c_p spike may cause wall temperature exceedance.
DQ-E	LOW	Cell assignment sensitivity: N_{nozzle} from 5 to 10 changes $\varepsilon_{\text{Mode2}}$ from 77 to 116.
DQ-F	LOW	Turbine efficiency at deep throttle; mitigated by variable-speed LOX pump.
DQ1	MED	Generator rotor dynamics at 20 000–40 000 RPM; gyroscopic coupling into engine gimbal; torsional vibration analysis of combined turbopump–generator shaft.
DQ2	MED	Inverter thermal management in vacuum; ≈ 25 kW dissipation requires dedicated radiator ($\approx 0.5 \text{ m}^2$ per engine).
DQ3	CLOSED	Deep-throttle power transition; fully resolved by dual-mode BVR at 4 kg (§8).
DQ4	MED	Bypass injector stability; substantially mitigated by S1+S3 (injection temperature 800 K→350 K).
DQ5	MED	NMC cell performance at cryogenic proximity ($< -20^\circ\text{C}$); active thermal sleeve required; 15–20 % capacity reduction.
DQ6	LOW	ROSA LILT qualification; moot if Psyche arrays selected.
DQ7	MED	HX-N Mode-2 bypass failsafe valve; must actuate in < 0.5 s on nozzle-loop shutdown to prevent thermal runaway.
DQ8	LOW	Psyche ZTJ LILT characterisation at 9.58 AU (Saturn/Titan); qualification verified to 2.9 AU only.
DQ-L3	HIGH	Cold start at 75 K (Enceladus); NMC battery loses 30–40 % capacity; active resistive heating required (standby < 5 W), adds mass.
DQ-L4	MED	H ₂ SO ₄ compatibility of XSLR-40 external components at Venus 56 km; Hastelloy C-276, PTFE, TiO ₂ coating qualification required.

12 Complete Parameter Tables

Table 12: Full XSLR-22 Config. D+ design-point parameters.

Parameter	Symbol	Value	Parameter	Symbol	Value
<i>Mission and scaling</i>			<i>Turbomachinery (Config. D+)</i>		
Sea-level thrust per engine	F_{SL}	22 100 N	Spike turbine power (avail.)	$\dot{W}_{turb,s}$	257 kW
Scaling factor	k	1.00	Nozzle turbine power (avail.)	$\dot{W}_{turb,n}$	45 kW
<i>Propellant flows</i>			Electric pump demand (shaft)	$\dot{W}_{pump,S}$	28.4 kW
Total mass flow	$\dot{m}_{eng}$	5.90 kg/s	Battery draw	P_{batt}	49 kW
LH ₂ mass flow	$\dot{m}_{H_2}$	1.017 kg/s	LOX pump demand	$\dot{W}_{pump,LOX}$	28 kW
LOX mass flow	$\dot{m}_{LOX}$	4.880 kg/s	Bypass surplus	$\dot{W}_{bypass}$	246 kW
<i>Geometry</i>			Power ratio (Config. D ref.)	Π_W	2.43
Total throat area	A_t	2.77 cm ²	<i>Performance</i>		
Cell throat half-height	h_t	7.50 mm	$I_{sp,vac}$	—	465 s
Plug length (80 %)	L_{plug}	37.6 cm	$I_{sp,SL}$	—	394 s
Engine span	W_{span}	37.5 cm	Throttle ratio	—	5:1
<i>Heat transfer</i>			Mode-2 ε_{eff}	—	96.7
Throat heat flux	q_0	17.04 MW/m ²	Mode-2 $I_{sp,vac}$	—	468.6 s
Nozzle loop heat load	$\dot{Q}_n$	154 kW	Battery mass	m_{batt}	36 kg
Spike loop heat load	$\dot{Q}_s$	992 kW	$T_{turb,max}$	—	700 K
Heat-load ratio	$\dot{Q}_s/\dot{Q}_n$	6.44			
<i>Coolant flows (design split $\alpha = 0.25$)</i>					
Spike design flow	$\dot{m}_{H_2,s}$	0.254 kg/s			
Nozzle design flow	$\dot{m}_{H_2,n}$	0.763 kg/s			
Spike outlet temperature	$T_{out,s}$	800 K			
Nozzle outlet temperature	$T_{out,n}$	81 K			
Nozzle turbine fraction	f_N	0.033			

Table 13: XSLR-220 Option C design-point additions over Config. D+.

Parameter	Symbol	Value	Parameter	Symbol	Value
Scaling factor	k	10.00	Bypass mass flow	$\dot{m}_{bypass}$	1.13 kg/s
Sea-level thrust per engine	F_{SL}	220 kN	S1+S3 injection temperature	T_{inj}	350 K
Total mass flow	$\dot{m}_{eng}$	59.00 kg/s	S1+S3 combined mass	—	20 kg
Spike turbine power (avail.)	$\dot{W}_{turb,s}$	2570 kW	Bootstrap battery	m_{batt}	35 kg
Generator extraction	$P_{extract}$	505 kW	BVR crossover RPL	RPL*	56.3 %
Chain efficiency	η_{chain}	0.562	BVR mass	—	4 kg
Residual bypass	$\dot{W}_{bypass}$	2065 kW	Generator + inverter mass	—	100–160 kg
Total addition over Config. D					147–205 kg

13 Key Equations Reference

Table 14: Complete equation reference for the XSLR architecture.

Quantity	Equation	Eq.
ISRU propulsion chain	$\text{H}_2\text{O} \rightarrow \text{WTP} \rightarrow \text{SEA} \rightarrow \text{XSLR} \rightarrow \Delta v$	(1)
Throat heat flux	$q_0 = C_B p_c^{0.8} T_c^{0.68}$	(3)
Nozzle cooling area	$A_{\text{cool},n} = k_{\text{Rao}} A_t$	(4)
Spike cooling area	$A_{\text{cool},s} = L_{\text{plug}} W_{\text{span}} k_{\text{spike}}$	(5)
Nozzle heat load	$\dot{Q}_n = q_0 A_{\text{cool},n} \bar{\eta}_n$	(6)
Spike heat load	$\dot{Q}_s = q_0 A_{\text{cool},s} \bar{\eta}_s$	(7)
Min coolant flows	$\dot{m}_{\text{min}} = \dot{Q}/(\bar{c}_p \Delta T_{\text{max}})$	(15)
Isentropic expansion eff.	$\eta_{\text{exp}} = 1 - \Pi_t^{-(\gamma-1)/\gamma}$	(17)
Turbopump power ratio	$\Pi_W = \dot{W}_{\text{turb,total}}/\dot{W}_{\text{pump,total}}$	(22)
Mode-2 area ratio	$\varepsilon_{\text{eff},\text{M2}} = (N_{\text{cell}}/N_{\text{spike}}) \varepsilon$	(23)
Electric pump demand	$\dot{W}_{\text{pump},s} = \dot{m}_s \Delta p/(\rho \eta_p)$	(27)
Battery draw	$P_{\text{batt}} = \dot{W}_{\text{pump},s}/(\eta_p \eta_m \eta_{\text{inv}})$	(28)
Nozzle turbine fraction	$f_N = \dot{Q}_n/[(T_{\text{out}} - T_{\text{in}}) \dot{m}_n \bar{c}_p]$	(29)
Battery mass	$m_{\text{batt}} = k_b \max(P_{\text{batt}}/\xi_b, W_l/\tau_b)$	(31)
Turbine efficiency	$\eta_t = \alpha_1 (u/C_{\text{ad}})^2 + \alpha_2 (u/C_{\text{ad}}) + \alpha_3$	(33)
Chain efficiency	$\eta_{\text{chain}} = \eta_{\text{gen}} \eta_{\text{inv}} \eta_{\text{mot}} \eta_p$	(36)
Generator extraction	$P_{\text{extract}} = \dot{W}_{\text{pump},s}/\eta_{\text{chain}}$	(37)
S1 recuperator duty	$\dot{Q}_{\text{HXR}} = \dot{m}_{\text{bypass}} c_p (T_{\text{in}} - T_{\text{out}})$	—
BVR crossover RPL	$\text{RPL}^* = \dot{W}_{\text{pump},s}/\dot{W}_{\text{gen},100\%}$	(41)
Battery recharge time	$t_{\text{rech}} = W_{\text{stored}}/(\eta_{\text{ch}} P_{\text{solar}})$	(42)
Aerospike I_{sp} at Titan	$I_{\text{sp,aero}} = \eta_{\text{CF}} I_{\text{sp,vac}}$	(45)
Min lander thrust	$F_{\text{vac,min}} = (T/W_{\text{req}}) m_0 g/\eta_{\text{CF}}$	(??)

14 Conclusions

This work consolidates the full XSLR design heritage into a single self-contained reference. Configuration D+, a parallel full-expander dual-loop aerospike augmented by a hybrid EEC, is the recommended baseline across all thrust classes:

- At **22 kN** (demonstrator): $I_{sp,vac} = 465$ s, throttle 5:1, Mode-2 + throttle (12–100 % RPL), battery 36 kg, $T_{turb} \leq 700$ K, trade-matrix score 0.760.
- At **40 kN** (planetary lander): mission-enabling at Titan (I_{sp} factor $\times 7.1$ over a bell nozzle, mass ratio 1.90 vs. 95.6); Venus, Enceladus all within budget. Extended D+ with 8:1 pintle throttle required for Titan precision landing (DQ-L1).
- At **220 kN** (vehicle engine): three non-invariances mandate turbogenerator Option C; bootstrap battery 35 kg; total electromechanical addition 147–205 kg per engine (vs. 618 kg worst-case battery for Option B). S1+S3 bypass management (20 kg) reduces injection temperature from 800 K to 350 K. DQ3 fully resolved by 4 kg BVR.
- At **250 kN** (extreme scale): Config. D intermediate hot-fire step for DQ-A/DQ-D; Option C is the required flight architecture.

Seventeen open design questions are consolidated (Table 11); three are high priority (DQ-A, DQ-D, DQ-L1); DQ3 is fully resolved. The single highest-priority experimental milestone is hot-fire validation of the bypass injector at ≥ 50 % bypass fraction, which simultaneously addresses DQ-A (spike heat load confirmation) and DQ-D (injector stability).

The recommended full-scale XSLR architecture is:

Config. D+ (parallel full-expander dual-loop, hybrid EEC) + Option C turbogenerator + S1+S3 bypass heat exchangers + BVR power controller + Psyche $\times 2$ solar arrays for in-orbit battery recharge. Total additions over Config. D baseline: 147–205 kg/engine (electromechanical) + 20 kg (HX) + 4 kg (BVR) + 285 kg (arrays, vehicle-level, shared).

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